

Factor Theory

Chapter Summary

Assets earn risk premiums because they are exposed to underlying factor risks. The capital asset pricing model (CAPM), the first theory of factor risk, states that assets that crash when the market loses money are risky and therefore must reward their holders with high risk premiums. While the CAPM defines bad times as times of low market returns, multifactor models capture multiple definitions of bad times across many factors and states of nature.

1. The 2008–2009 Financial Crisis

During the financial crisis of 2008 and 2009, the price of most risky assets plunged. Table 6.1 shows that U.S. large cap equities returned –37%; international and emerging markets equities had even larger losses. The riskier fixed income securities, like corporate bonds, emerging market bonds, and high yield bonds, also fell, tumbling along with real estate. “Alternative” investments like hedge funds, which trumpeted their immunity to market disruptions, were no safe refuge: equity hedge funds and their fixed income counterparts fell approximately 20%. Commodities had losses exceeding 30%. The only assets to go up during 2008 were cash (U.S. Treasury bills) and safe-haven sovereign bonds, especially long-term U.S. Treasuries.

Why did so many asset classes crash all at once? And given that they did, was the concept of diversification dead?

In this chapter, we develop a theory of factor risk premiums. The factor risks constitute different flavors of *bad times* and the investors who bear these factor risks need to be compensated in equilibrium by earning factor risk premiums. Assets have risk premiums not because the assets themselves earn risk premiums; assets are bundles of factor risks, and it is the exposures to the underlying factor risks that earn risk premiums. These factor risks manifest during bad times such as the financial crisis in late 2008 and early 2009.